

Iron Microplate Data Analysis Workflow

When collecting data from microplate reader:

- Paste data into the sample data worksheet (shown below)
- Fill in sample IDs and Dilution information as well

The screenshot shows an Excel spreadsheet titled "FE_20230103_Std1". The spreadsheet is divided into several tabs: "Std1", "Samples1", "Samples2", "Samples3", "Samples4", "Samples5", "Samples6", "Std2", and "FE_20230103_Samples". The "Std1" tab is active, showing a table with columns for standards (A-H) and rows for various parameters (1-12). The "Samples1" tab is also visible, showing a table with columns for samples (1-12) and rows for various parameters (1-12). The "FE_20230103_Samples" tab is also visible, showing a table with columns for samples (1-12) and rows for various parameters (1-12).

Std1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
1	Standards1																												
2																													
3	stds1	1	2	3	4	5	6	7	8	9	10	11	12																
4	A	0.204	0.262	0.204	0.397	0.344	0.317	0.041	0.041	0.042	0.039	0.039	0.04	Hz															
5	B	0.192	0.192	0.192	0.192	0.192	0.192	0.041	0.041	0.041	0.041	0.041	0.041	Hz															
6	C	0.131	0.141	0.135	1.333	1.355	1.336	0.039	0.041	0.04	0.049	0.058	0.051	Hz															
7	D	0.091	0.094	0.088	0.052	0.049	0.06	0.039	0.039	0.039	0.048	0.053	0.049	Hz															
8	E	0.065	0.069	0.062	0.085	0.098	0.049	0.037	0.038	0.038	0.054	0.06	0.05	Hz															
9	F	0.055	0.053	0.054	0.049	0.048	0.048	0.038	0.038	0.038	0.05	0.056	0.049	Hz															
10	G	0.047	0.047	0.046	0.065	0.047	0.059	0.039	0.039	0.039	0.076	0.067	0.06	Hz															
11	H	0.038	0.039	0.038	0.038	0.037	0.037	0.037	0.037	0.037	0.036	0.037	0.037	Hz															
12																													
13																													
14																													
15																													
16																													
17																													
18	A	0.309	0.342	0.316	0.532	0.503	0.448	0.169	0.173	0.181	0.267	0.218	0.256	Hz															
19	B	0.251	0.25	0.237	0.995	0.939	0.857	0.136	0.134	0.139	0.366	0.361	0.296	Hz															
20	C	0.171	0.198	0.185	1.408	1.412	1.373	0.105	0.095	0.112	0.04	0.04	0.04	Hz															
21	D	0.119	0.123	0.117	0.041	0.04	0.04	0.084	0.081	0.084	0.04	0.04	0.04	Hz															
22	E	0.078	0.083	0.079	0.041	0.04	0.04	0.057	0.04	0.06	0.039	0.039	0.04	Hz															
23	F	0.065	0.063	0.064	0.04	0.04	0.04	0.051	0.052	0.052	0.04	0.04	0.039	Hz															
24	G	0.052	0.056	0.052	0.04	0.04	0.04	0.045	0.046	0.046	0.04	0.04	0.04	Hz															
25	H	0.04	0.039	0.04	0.039	0.038	0.037	0.039	0.038	0.038	0.037	0.037	0.037	Hz															
26	Date	Plate	Well Num	Concentr	dilution	fr	Abs1																						
27	20230103	Standards1	1	Fe(II) A	1	0.30	0.31																						
28	20230103	Standards1	2	Fe(II) A	1	0.26	0.34																						
29	20230103	Standards1	3	Fe(II) A	1	0.20	0.32																						
30	20230103	Standards1	4	Fe(II) 100	1	0.40	0.53																						
31	20230103	Standards1	5	Fe(II) 100	1	0.34	0.50																						
32	20230103	Standards1	6	Fe(II) 100	1	0.32	0.45																						
33	20230103	Standards1	7	Fe(III) A	1	0.04	0.17																						
34	20230103	Standards1	8	Fe(III) A	1	0.04	0.17																						
35	20230103	Standards1	9	Fe(III) A	1	0.04	0.18																						
36	20230103	Standards1	10	Fe(III) 100	1	0.04	0.27																						

The pasted/input data will fill the columns below that can be copied over to a new sheet

- Copy standards over into a new sheet, you can copy all standards into one sheet for one day
- Copy samples over into a new, but separate sheet. Put all samples from that day on the same sheet.

AutoSave FE_20230103_Std1 Wilson, Stephanie

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H122 0.046

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Date	Plate	Well Num	ID	dilution	fa	Abs1og	Abs2og	Abs1	Abs2									
2	20230103	Standards1	1	Fe(II) A	1	0.204	0.309	0.204	0.309										
3	20230103	Standards1	2	Fe(II) A	1	0.262	0.342	NA	NA										
4	20230103	Standards1	3	Fe(II) A	1	0.204	0.316	0.204	0.316										
5	20230103	Standards1	4	Fe(II) 100	1	0.397	0.532	0.397	0.532										
6	20230103	Standards1	5	Fe(II) 100	1	0.344	0.503	0.344	0.503										
7	20230103	Standards1	6	Fe(II) 100	1	0.317	0.448	0.317	0.448										
8	20230103	Standards1	7	Fe(III) A	1	0.041	0.169	0.041	0.169										
9	20230103	Standards1	8	Fe(III) A	1	0.041	0.173	0.041	0.173										
10	20230103	Standards1	9	Fe(III) A	1	0.042	0.181	0.042	0.181										
11	20230103	Standards1	10	Fe(III) 100	1	0.039	0.267	0.039	0.267										
12	20230103	Standards1	11	Fe(III) 100	1	0.039	0.218	0.039	0.218										
13	20230103	Standards1	12	Fe(III) 100	1	0.04	0.256	0.04	0.256										
14	20230103	Standards1	13	Fe(II) B	1	0.192	0.251	0.192	0.251										
15	20230103	Standards1	14	Fe(II) B	1	0.192	0.25	0.192	0.25										
16	20230103	Standards1	15	Fe(II) B	1	0.156	0.237	NA	NA										
17	20230103	Standards1	16	Fe(II) 250	1	0.801	0.995	0.801	0.995										
18	20230103	Standards1	17	Fe(II) 250	1	0.784	0.939	0.784	0.939										
19	20230103	Standards1	18	Fe(II) 250	1	0.697	0.857	NA	NA										
20	20230103	Standards1	19	Fe(III) B	1	0.042	0.136	0.042	0.136										
21	20230103	Standards1	20	Fe(III) B	1	0.041	0.134	0.041	0.134										
22	20230103	Standards1	21	Fe(III) B	1	0.041	0.139	0.041	0.139										
23	20230103	Standards1	22	Fe(III) 500	1	0.048	0.366	0.048	0.366										
24	20230103	Standards1	23	Fe(III) 500	1	0.047	0.361	0.047	0.361										

FE_20230103_Std1 Stds Samples1 Samples2 Samples3 Samples4 Samples5 Samples6 Stds2 FE_20230103_Samples

Average: 0.04675 Count: 8 Sum: 0.374

2:09 PM 1/17/2023

- This is an example of a standards sheet that is saved as a csv to a folder from that analysis day – In this case some of the standard replicates were not great in this case I have made the original data Abs1og and Abs2 og (og = original) and then for samples I want to remove I have input NA in the new Abs1 and Abs2 columns.

AutoSave FE_20230103_Std1 Wilson, Stephanie

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J4

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Date	Plate	Well Num	SampleID	dilution	fa	Abs1	Abs2								
2	20230103	Samples1	1	stds	1	1.138	1.315									
3	20230103	Samples1	2	stds	1	1.312	1.401									
4	20230103	Samples1	3	stds	1	1.188	1.472									
5	20230103	Samples1	4	270	1	0.055	0.04									
6	20230103	Samples1	5	270	1	0.051	0.039									
7	20230103	Samples1	6	270	1	0.053	0.041									
8	20230103	Samples1	7	278	1	0.066	0.094									
9	20230103	Samples1	8	278	1	0.056	0.078									
10	20230103	Samples1	9	278	1	0.074	0.1									
11	20230103	Samples1	10	284	7	0.048	0.052									
12	20230103	Samples1	11	284	7	0.048	0.053									
13	20230103	Samples1	12	284	7	0.048	0.055									
14	20230103	Samples1	13	stds	1	0.038	0.039									
15	20230103	Samples1	14	stds	1	0.038	0.039									
16	20230103	Samples1	15	stds	1	0.038	0.039									
17	20230103	Samples1	16	271	7	1.724	1.785									
18	20230103	Samples1	17	271	1	1.745	1.775									
19	20230103	Samples1	18	271	1	1.575	1.694									
20	20230103	Samples1	19	279	7	0.194	0.297									

FE_20230103_Std1 Stds Samples1 Samples2 Samples3 Samples4 Samples5 Samples6 Stds2 FE_20230103_Samples

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- This is an example of a samples sheet that is saved as a csv to a folder from that analysis day.

Next step: